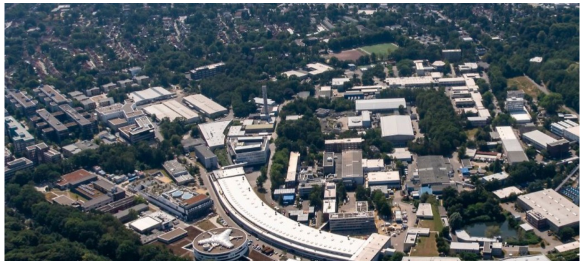
MXCuBE site report

P11, DESY (Hamburg, Germany)

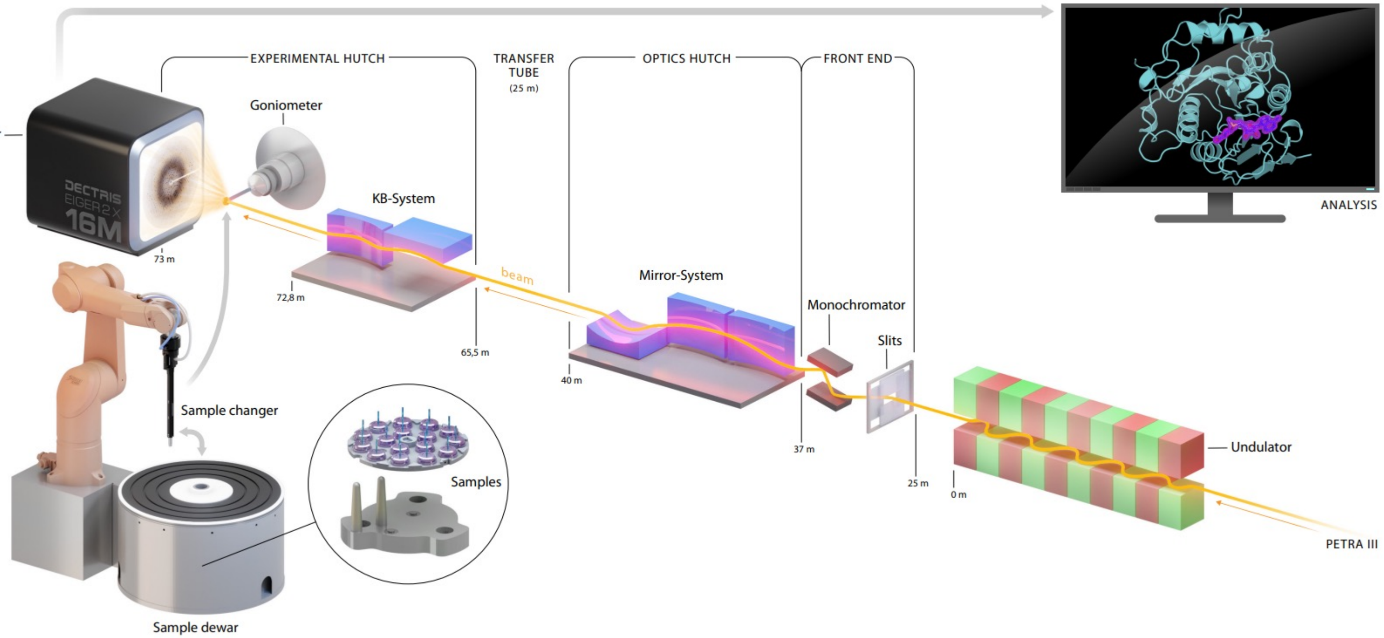
Brief beamline summary

The High-throughput Macromolecular Crystallography Beamline P11



Crystallography Experiments

- In user operation since 2013
- Broad energy range: 5.5 - 28 keV
- High-speed sample changer with capacity (23 unipucks = 368 samples)
- High precision single axis goniometer: 0.0001° at 120°/s
- Eiger2 X 16M
- XRF for experimental phasing (SAD/MAD) and element analysis



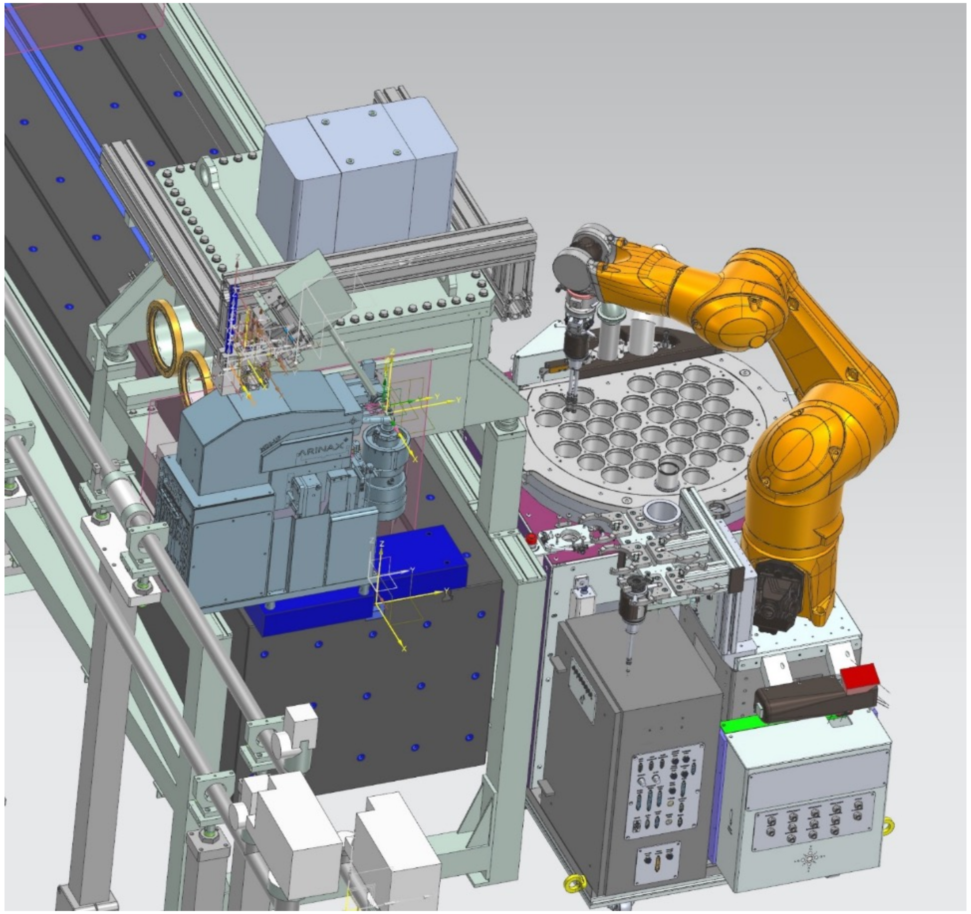
MXCuBE

- mxcubecore **2.5.0** (develop branch)
- Latest mxcubeqt (develop branch) with a local changes
- Python 3.11.9, PyQt 5.15.10.

Developments since last meeting

- Rebased local mxcubecore to the “shutdown-latest” version
- Adapted the new LIMS component
- Moved to YAML config
- Made RUFF happy with DESY components (pre-commit hook)
- Qt branch has been changed locally to make it work with the latest updates
- Extended the training dataset of P11 images for Murko
- Preparations for the new hardware

Beamline hardware upgrade



PETRA III autumn shutdown

The diffractometer ARINAX MD3-up
- Multi-axis goniometry
- Improved grid scans
- Helical scans

The FLEX-ED37 sample changer
- Increased sample capacity
- faster sample mount cycle

Overall aim: gradual replacement of the aging instruments and improving user-operation reliability

Plans for the next six months

- Preparation for the installation of new hardware
- X-Ray centering + mesh-scans
- ISPyB/EXI EDNA Workflows display
- Optimize the usage of murko-centering
- Code cleanup
- New PRs